1 Introduction to Statistics

Having developed a sufficient foundation in probability, we are now ready to apply it to study statistics. We first start by defining a simple data model to analyze.

1.1 Random Samples

Data collected in an experiment represents several observations on a variable of interest. Our focus will be on one type of collecting data called random sampling.

Definition 1 (Random Samples)**.** The random variables $X_1, \dots, X_n$ are called a random sample of size n from the population F if $X_1, \dots, X_n$ independent and identically distributed random variables with CDF F.

Our goal is to study facts about the population F using many realizations of it. Random sampling involve independent trials of the same experiment. To simplify our problem, suppose that we know the general behavior of the population, but we are unsure of precise behavior encoded by a parameter θ. The goal is to learn θ from our samples.

1.1.1 Types of Random Samples

If we do not want to ask the same person twice in a survey, then each draw removes someone from a finite population, and hence changes the remaining population for the next draw. This is sampling without replacement, and the resulting samples are identically distributed (Problem 1.1) but *not* independent.

Definition 2 (Sampling without replacement)**.** Sampling without replacement follows the following rule. Let $\{x_1, \dots, x_N\}$ be a finite population and suppose we select one element from that set uniformly at random. Let X_1 denote the value of this first draw. The element X_1 is removed from the set, and we sample X_2 from the remaining $N-1$ elements uniformly at random, and so on.

Sampling without replacement from a finite population is sometimes called *simple random sampling*. A less realistic, but much simpler, way of sampling is sampling with replacement.

Definition 3 (Sampling with replacement)**.** Sampling with replacement follows the following rule. Let $\{x_1, \dots, x_N\}$ be a finite population and suppose we select one element from that set uniformly at random. Let X_1 denote the value of this first draw. The element X_1 is put back in the set, and we sample X_2 from all N elements uniformly at random, and so on.

Sampling with replacement produces a random sample in the sense of Definition 1, where F is the uniform distribution on $\{x_1, \dots, x_N\}$. A random sample is often described as sampling from an *infinite population*, because when the population is large, removing one element has almost no effect on the remaining elements, so both types of sampling are nearly equivalent. Or alternatively, it is hard to sample the same person twice in a very large population.

Proposition 1

Consider a population $\{x_1, \dots, x_N\}$ of N elements. Suppose that we want to sample K elements. Let $(X_1, \dots, X_K)$ and $(Y_1, \dots, Y_K)$ denote samples obtained via sampling without replacement and sampling with replacement respectively. For every (Borel) set $A \subseteq \mathbb{R}^K$, we have

$$\big|\, \mathbb{P}((X_1, \dots, X_K) \in A) - \mathbb{P}((Y_1, \dots, Y_K) \in A) \big| \leq \frac{K(K-1)}{2N}.$$

In particular, for a fixed sample size K, the two sampling schemes become indistinguishable as $N \to \infty$.

Throughout the rest of the course, we will work with large populations, so we will work with random samples, without losing so much precision (see Problem 1.3 for a numerical comparison).

1.2 Summary Statistics

We are often not interested in the entire sample, but a summary of the sample called a statistic.

Definition 4 (Statistic)**.** Let $X_1, \ldots, X_n$ be a random sample of size n from a population and let $T(x_1, \ldots, x_n)$ be a real-valued (non-random) function (or vector). The random variable $Y = T(X_1, \ldots, X_n)$ is called the *statistic* and the distribution of Y is called the *sampling distribution*.

In statistics, our main goal is often to understand the *sampling distribution* because it will tell us if the observed summary is expected or rare. The function T can be very general, but it cannot depend on an unknown parameter of the model. A statistic must be something we can compute from the data alone.

Example 1. Let $X_1, \ldots, X_n$ be a random sample from $\mathrm{N}(\mu, \sigma^2)$ where μ and σ^2 are unknown. Then

$$\sum_{i=1}^{n} X_i, \qquad \max_{i \le n} X_i, \qquad \sum_{i=1}^{n} X_i^2$$

are statistics. However, $\frac{X_1 - \mu}{\sigma}$ is not a statistic since we cannot compute it without knowing μ and σ.

1.2.1 Examples of Common Statistics

The moments are the idealized summaries of a random variable, so two of the most common statistics are the sample analogues of the mean and variance of the population.

Definition 5 (Sample Mean)**.** The sample mean is the average of the sample

$$\bar{X} = \frac{1}{n} \sum_{i=1}^{n} X_i.$$

By the central limit theorem, if the population has finite variance σ^2, then the sampling distribution of $\bar{X}$ is approximately $\mathrm{N}(\mu, \frac{\sigma^2}{n})$ for large n, no matter what the population is. In Section 1.3, we compute the sampling distribution exactly in some important cases.

Definition 6 (Sample Variance)**.** The sample variance is the variance of the sample

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})^2.$$

The sample standard deviation is $S = \sqrt{S^2}$.

It might be somewhat unexpected that we are normalizing by $\frac{1}{n-1}$ instead of $\frac{1}{n}$. We will see in Theorem 1 that this choice is exactly what makes $\mathbb{E}[S^2] = \sigma^2$.

The following algebraic identities hold for any list of numbers, and they make computations with $\bar{X}$ and S^2 much easier.

Proposition 2

Let $x_1, \ldots, x_n$ be any real numbers, and let $\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$ and $s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2$. Then

1. $\min_{a \in \mathbb{R}} \sum_{i=1}^{n} (x_i - a)^2 = \sum_{i=1}^{n} (x_i - \bar{x})^2$, *and the minimum is attained only at $a = \bar{x}$.*
2. $(n-1)s^2 = \sum_{i=1}^{n} (x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - n\bar{x}^2$.

Remark 1. The first identity is the sample analogue of the fact that the expected value minimizes the mean squared error, and the second is the sample analogue of $\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$.

We now compute the expected values of the two statistics. The first two parts were already used in the proof of the weak law of large numbers.

Theorem 1 (*Moments of the Sample Mean and Variance*)

Let $X_1, \ldots, X_n$ be a random sample from a population with mean μ and finite variance σ^2. Then

1. $\mathbb{E}[\bar{X}] = \mu$,
2. $\mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n}$,
3. $\mathbb{E}[S^2] = \sigma^2$.

Remark 2. A statistic whose expected value is the parameter it is trying to estimate is called *unbiased*, and we will study this notion carefully when we cover point estimation. If we instead normalize by $\frac{1}{n}$, then the expected value is $\frac{n-1}{n}\sigma^2 < \sigma^2$ (Problem 1.5). The intuition is that we measure the deviations $X_i - \bar{X}$ from $\bar{X}$ instead of from μ. By Proposition 2, $\bar{X}$ is the point closest to the data, so these deviations are systematically too small. Dividing by $n-1$ exactly corrects for this. Another way to see this is that the n deviations satisfy one constraint $\sum_{i=1}^{n}(X_i - \bar{X}) = 0$, so only $n-1$ of them are free. We will make this *degrees of freedom* heuristic precise in Theorem **??**.

1.3 Distribution of Sums from a Random Sample

Theorem 1 tells us the center and spread of $\bar{X}$, but not its full sampling distribution. When the MGF of the population exists, the independence property of MGFs gives us a simple way to identify it.

Theorem 2 (*MGF of the Sample Mean*)

Let $X_1, \ldots, X_n$ be a random sample from a population with MGF $M_X(t)$. Then the MGF of the sample mean is

$$M_{\bar{X}}(t) = \big[M_X(t/n)\big]^n.$$

If the MGF does not exist, or the resulting MGF is not one we recognize, then we can compute the distribution of a sum directly.

Theorem 3 (*Convolution Formula*)

Let X and Y be independent continuous random variables with PDFs f_X and f_Y. Then $Z = X+Y$ is a continuous random variable with PDF

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(w) f_Y(z-w)\, dw.$$

Remark 3. The sample mean does not always become more concentrated as n grows. If $X_1, \ldots, X_n$ is a random sample from the Cauchy distribution with PDF $f(x) = \frac{1}{\pi(1+x^2)}$, then one can use the convolution formula (see Casella–Berger, Example 5.2.10) to show that $\bar{X}$ also has the same Cauchy distribution for every n. Averaging 1000 observations gives no improvement over a single observation! This does not contradict the law of large numbers or the CLT, since the Cauchy distribution does not have a finite mean.

1.4 The Gamma Function and the Gamma Distribution

Many of the distributions in this chapter, including the exponential and chi-squared distributions and the normalizing constants of the t and F distributions, are built out of one special function. It generalizes the factorial to non-integer values.

Definition 7 (Gamma Function)**.** For $\alpha > 0$, the gamma function is

$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x}\, dx.$$

The integral converges for every $\alpha > 0$. Near 0 the integrand behaves like $x^{\alpha-1}$, which is integrable since $\alpha > 0$, and near ∞ the exponential decay beats any power of x.

Proposition 3 (*Properties of the Gamma Function*)

1. $\Gamma(\alpha+1) = \alpha\Gamma(\alpha)$ *for all* $\alpha > 0$.
2. $\Gamma(n) = (n-1)!$ *for every positive integer* n.
3. $\Gamma(\frac{1}{2}) = \sqrt{\pi}$.

Remark 4. Applying the first property repeatedly and then the third, we can compute the gamma function at half-integers,

$$\Gamma\Big(k+\frac{1}{2}\Big) = \Big(k-\frac{1}{2}\Big)\Big(k-\frac{3}{2}\Big)\cdots\frac{1}{2}\,\Gamma\Big(\frac{1}{2}\Big) = \frac{(2k-1)!!}{2^k}\sqrt{\pi}.$$

For example, $\Gamma(\frac{3}{2}) = \frac{\sqrt{\pi}}{2}$ and $\Gamma(\frac{5}{2}) = \frac{3\sqrt{\pi}}{4}$. The double factorial here is the same one that appears in the normal moments $\mathbb{E}[Z^{2k}] = (2k-1)!!$ (Problem **??**). For large α, the gamma function grows like the factorial, and Stirling's formula says that $\Gamma(\alpha+1) \approx \sqrt{2\pi\alpha}\,(\frac{\alpha}{e})^\alpha$ as $\alpha \to \infty$.

The gamma function is exactly the normalizing constant of the following family of distributions.

Definition 8 (Gamma Distribution)**.** A random variable X has a gamma distribution with shape $\alpha > 0$ and scale $\beta > 0$, denoted $X \sim \Gamma(\alpha, \beta)$, if it has PDF

$$f_X(x) = \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta}, \qquad x > 0.$$

Remark 5. We use the *scale* parametrization, which matches Casella–Berger and our notation $\mathrm{Exp}(\theta)$ for the exponential distribution with mean θ. Many other sources use the *rate* parametrization instead, where the PDF is $\frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}$ with rate $\lambda = \frac{1}{\beta}$. Always check which convention a source uses before copying a formula for the mean or the MGF.

The shape parameter α determines what the PDF looks like, while the scale β only stretches or compresses it (see Proposition 5). If $\alpha < 1$, then the PDF blows up at 0 and is decreasing, if $\alpha = 1$, then it is the exponential PDF, and if $\alpha > 1$, then it has a single peak at $x = (\alpha-1)\beta$. As α grows, the PDF looks more and more like a normal PDF, which is consistent with the central limit theorem since $\Gamma(n, \beta)$ is the distribution of a sum of n i.i.d. $\mathrm{Exp}(\beta)$ random variables (Proposition 5).

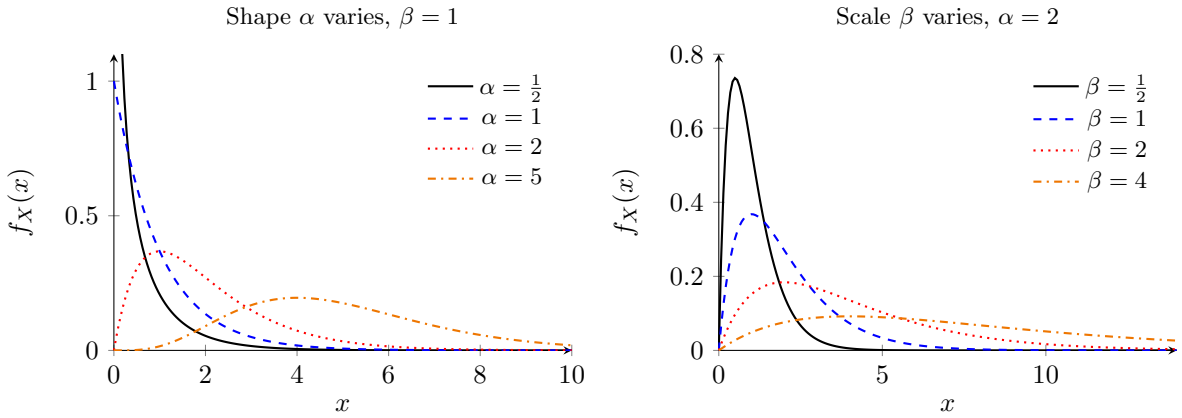


Figure: The PDF of the $\Gamma(\alpha, \beta)$ distribution. Changing the shape α changes what the PDF looks like (left), while changing the scale β only stretches it horizontally and compresses it vertically (right).

Example 2. Several distributions we have seen are gamma distributions.

1. When $\alpha = 1$, the PDF is $\frac{1}{\beta}e^{-x/\beta}$, so $\Gamma(1, \beta) = \text{Exp}(\beta)$ is the exponential distribution with mean β.
2. When $\alpha = n$ is an integer, $\Gamma(n, \beta)$ is the waiting time until the nth arrival when the times between arrivals are i.i.d. $\text{Exp}(\beta)$ (Problem 1.11).

Proposition 4 (*Moments and MGF of the Gamma Distribution*)

Let $X \sim \Gamma(\alpha, \beta)$.

1. *For any $k > -\alpha$,*
$$\mathbb{E}[X^k] = \frac{\Gamma(\alpha + k)}{\Gamma(\alpha)}\beta^k.$$
2. *$\mathbb{E}[X] = \alpha\beta$ and $\text{Var}(X) = \alpha\beta^2$.*
3. *The MGF is $M_X(t) = (1 - \beta t)^{-\alpha}$ for $t < \frac{1}{\beta}$.*

The MGF makes it easy to see how the gamma distribution behaves under scaling and sums.

Proposition 5 (*Scaling and Sums of Gamma Random Variables*)

1. *If $X \sim \Gamma(\alpha, \beta)$ and $c > 0$, then $cX \sim \Gamma(\alpha, c\beta)$.*
2. *If $X_1, \ldots, X_k$ are independent and $X_i \sim \Gamma(\alpha_i, \beta)$ with the same scale β, then*
$$X_1 + \cdots + X_k \sim \Gamma(\alpha_1 + \cdots + \alpha_k, \beta).$$

Remark 6. In particular, if $X_1, \ldots, X_n$ are i.i.d. $\text{Exp}(\theta)$, then $\sum_{i=1}^n X_i \sim \Gamma(n, \theta)$, and the additivity of the chi-squared distribution (Lemma **??**) is the special case $\beta = 2$. The scales must match, since otherwise the MGFs do not combine into a single gamma MGF.

1.5 Example Problems

Problem 1.1. Let $X_1, \ldots, X_N$ be samples from a population $\{x_1, \ldots, x_N\}$ of distinct values without replacement. What is the marginal distribution of X_i?

Solution 1.1. We have that

$$\mathbb{P}(X_1 = x) = \frac{1}{N}$$

for each $x \in \{x_1, \dots, x_N\}$. Since the second sample depends on the outcome of the first, we need to condition on the outcome of the first draw

$$\mathbb{P}(X_2 = x) = \sum_{i=1}^{N} \mathbb{P}(X_2 = x, X_1 = x_i) = \sum_{i=1}^{N} \mathbb{P}(X_2 = x \mid X_1 = x_i)\, \mathbb{P}(X_1 = x_i).$$

If $x_i = x$, then $\mathbb{P}(X_2 = x \mid X_1 = x_i) = 0$, but for all others it is $\frac{1}{N-1}$. Therefore,

$$\mathbb{P}(X_2 = x) = (N-1)\frac{1}{N-1}\frac{1}{N} = \frac{1}{N}.$$

A similar computation shows that

$$\begin{aligned}\mathbb{P}(X_3 = x) &= \sum_{j=1}^{N}\sum_{i=1}^{N} \mathbb{P}(X_3 = x, X_2 = x_j, X_1 = x_i) \\ &= \sum_{j=1}^{N}\sum_{i=1}^{N} \mathbb{P}(X_3 = x \mid X_2 = x_j, X_1 = x_i)\, \mathbb{P}(X_2 = x_j \mid X_1 = x_i)\, \mathbb{P}(X_1 = x_i).\end{aligned}$$

The only non-zero terms are the $(N-1)(N-2)$ pairs (i, j) where x_i, x_j, x are distinct, and each of these terms is equal to $\frac{1}{N-2}\frac{1}{N-1}\frac{1}{N}$. Therefore,

$$\mathbb{P}(X_3 = x) = (N-1)(N-2)\frac{1}{N-2}\frac{1}{N-1}\frac{1}{N} = \frac{1}{N},$$

and so on.

Alternative Solution: Instead of looking at the problem sequentially, we can look at the problem as a whole. We can denote the outcome of a random sample as a uniform permutation of $(x_1, \dots, x_N)$. The order of the permutation determines the outcomes of $X_1, X_2, \dots, X_N$. This is clear by symmetry. Then for a fixed value $x \in \{x_1, \dots, x_N\}$, we have

$$\mathbb{P}(X_j = x) = \frac{1}{N}$$

since x can appear in any of the N spots with the same probability.

Problem 1.2. Let $\{x_1, \dots, x_N\}$ be a population with mean and variance

$$\mu = \frac{1}{N}\sum_{k=1}^{N} x_k \qquad \text{and} \qquad \sigma^2 = \frac{1}{N}\sum_{k=1}^{N}(x_k - \mu)^2.$$

Let $X_1, \dots, X_n$ be a sample of size $n \le N$ drawn without replacement, and let $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$.

1. Show that $\operatorname{Cov}(X_i, X_j) = -\frac{\sigma^2}{N-1}$ for $i \neq j$.
2. Show that $\mathbb{E}[\bar{X}] = \mu$ and
$$\operatorname{Var}(\bar{X}) = \frac{\sigma^2}{n} \cdot \frac{N-n}{N-1}.$$

Solution 1.2.

Part 1: By Problem 1.1, each X_i is uniform on the population, so $\mathbb{E}[X_i] = \mu$ and $\mathrm{Var}(X_i) = \sigma^2$. By the alternative solution of Problem 1.1, the full sample $(X_1, \dots, X_N)$ is a uniformly random permutation of the population, so every pair (X_i, X_j) with $i \neq j$ has the same joint distribution. Therefore $\mathrm{Cov}(X_i, X_j) = c$ for some constant c that does not depend on $i \neq j$.

To find c, we use a trick. If we sample the entire population, then $\sum_{i=1}^N X_i = \sum_{k=1}^N x_k$ is not random, so its variance is zero. By the formula for the variance of linear combinations (Week 5),

$$0 = \mathrm{Var}\left(\sum_{i=1}^N X_i\right) = \sum_{i=1}^N \mathrm{Var}(X_i) + \sum_{i \neq j} \mathrm{Cov}(X_i, X_j) = N\sigma^2 + N(N-1)c \implies c = -\frac{\sigma^2}{N-1}.$$

Part 2: By linearity, $\mathbb{E}[\bar{X}] = \frac{1}{n}\sum_{i=1}^n \mathbb{E}[X_i] = \mu$. Since the first n draws of the full sample are a sample of size n without replacement, Part 1 implies that

$$\mathrm{Var}(\bar{X}) = \frac{1}{n^2}\left(n\sigma^2 + n(n-1)\left(-\frac{\sigma^2}{N-1}\right)\right) = \frac{\sigma^2}{n}\left(1 - \frac{n-1}{N-1}\right) = \frac{\sigma^2}{n} \cdot \frac{N-n}{N-1}.$$

Remark 7. The factor $\frac{N-n}{N-1}$ is called the *finite population correction*. The draws are negatively correlated since removing a large value makes the remaining values smaller on average, so the sample mean is less variable than for a random sample. If n is small compared to N, then the correction is close to 1 and we recover $\mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n}$ from Theorem 1, which is consistent with Proposition 1. If $n = N$, then $\mathrm{Var}(\bar{X}) = 0$ since we have observed the entire population.

Problem 1.3. Suppose that $\{1, 2, \dots, 1000\}$ is a finite population, and we draw a sample of size 10 without replacement. What is the probability that all ten sample values are greater than 200? Compare this with the answer we would get if we pretend the sample is a random sample (i.e. sampling with replacement).

Solution 1.3. **Exact:** Sampling without replacement, the set of drawn values is equally likely to be any of the $\binom{1000}{10}$ subsets of size 10. There are $\binom{800}{10}$ subsets with all values greater than 200, so

$$\mathbb{P}(X_1 > 200, \dots, X_{10} > 200) = \frac{\binom{800}{10}}{\binom{1000}{10}} = \frac{800}{1000} \cdot \frac{799}{999} \cdots \frac{791}{991} = 0.106164.$$

Random sample approximation: If the X_i were independent with the same marginal distribution, then

$$\mathbb{P}(X_1 > 200, \dots, X_{10} > 200) = \mathbb{P}(X_1 > 200) \cdots \mathbb{P}(X_{10} > 200) = \left(\frac{800}{1000}\right)^{10} = 0.107374.$$

The difference is 0.0012, which is well within the guarantee $\frac{K(K-1)}{2N} = \frac{90}{2000} = 0.045$ from Proposition 1. Each factor in the exact product is very close to $\frac{800}{1000}$, which is why the approximation is so good.

Problem 1.4. Suppose the lifetimes (in years) of n light bulbs are a random sample from the $\mathrm{Exp}(\theta)$ distribution with mean θ.

1. What is the joint PDF of the sample?
2. What is the probability that all of the bulbs last longer than 2 years?

Solution 1.4. This problem follows by independence.

Part 1: The PDF of the population is $f(x \mid \theta) = \frac{1}{\theta} e^{-x/\theta}$ for $x > 0$, so indpendence,

$$f(x_1, \ldots, x_n \mid \theta) = \prod_{i=1}^{n} \frac{1}{\theta} e^{-x_i/\theta} = \frac{1}{\theta^n} e^{-\frac{1}{\theta}\sum_{i=1}^n x_i}, \qquad x_1, \ldots, x_n > 0.$$

Part 2: Since the joint PDF factorizes, the n-dimensional integral splits into a product of n one-dimensional integrals,

$$\begin{aligned}\mathbb{P}(X_1 > 2, \ldots, X_n > 2) &= \int_2^\infty \cdots \int_2^\infty \prod_{i=1}^{n} \frac{1}{\theta} e^{-x_i/\theta}\, dx_1 \cdots dx_n \\ &= \prod_{i=1}^{n} \int_2^\infty \frac{1}{\theta} e^{-x_i/\theta}\, dx_i = \left(e^{-2/\theta}\right)^n = e^{-2n/\theta}.\end{aligned}$$

Problem 1.5. Let $X_1, \ldots, X_n$ be a random sample from a population with mean μ and finite variance σ^2. Compute the expected value of

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^{n} (X_i - \bar{X})^2.$$

Solution 1.5. Notice that $\hat{\sigma}^2 = \frac{n-1}{n} S^2$, so by Theorem 1,

$$\mathbb{E}[\hat{\sigma}^2] = \frac{n-1}{n}\, \mathbb{E}[S^2] = \frac{n-1}{n} \sigma^2.$$

On average, $\hat{\sigma}^2$ underestimates σ^2, although the difference disappears as $n \to \infty$.

Problem 1.6. Let $X_1, \ldots, X_n$ be a random sample from the Ber(p) distribution.

1. Show that $S^2 = \frac{n}{n-1} \bar{X}(1 - \bar{X})$.
2. Verify directly that $\mathbb{E}[S^2] = p(1-p)$.

Solution 1.6. We apply the sum identities in Proposition 2.

Part 1: Since $X_i \in \{0, 1\}$, we have $X_i^2 = X_i$, so $\sum_{i=1}^n X_i^2 = n\bar{X}$. By Proposition 2,

$$(n-1)S^2 = \sum_{i=1}^{n} X_i^2 - n\bar{X}^2 = n\bar{X} - n\bar{X}^2 = n\bar{X}(1 - \bar{X}).$$

Part 2: Since $\mathbb{E}[\bar{X}] = p$ and $\mathrm{Var}(\bar{X}) = \frac{p(1-p)}{n}$, we have $\mathbb{E}[\bar{X}^2] = \mathrm{Var}(\bar{X}) + \mathbb{E}[\bar{X}]^2 = \frac{p(1-p)}{n} + p^2$. Therefore,

$$\mathbb{E}[\bar{X}(1 - \bar{X})] = p - \frac{p(1-p)}{n} - p^2 = p(1-p)\left(1 - \frac{1}{n}\right) = \frac{n-1}{n} p(1-p),$$

so $\mathbb{E}[S^2] = \frac{n}{n-1} \cdot \frac{n-1}{n} p(1-p) = p(1-p)$, as predicted by Theorem 1.

Remark 8. The "plug-in" estimate $\bar{X}(1 - \bar{X})$ of the population variance $p(1-p)$ is biased, which is another instance of Problem 1.5.

Problem 1.7. Let $X_1, \dots, X_n$ be a random sample from the $\Gamma(\alpha, \beta)$ distribution with PDF

$$f(x) = \frac{x^{\alpha-1}e^{-x/\beta}}{\Gamma(\alpha)\beta^\alpha}, \qquad x > 0.$$

By Proposition 4, the MGF is $M_X(t) = (1-\beta t)^{-\alpha}$ for $t < \frac{1}{\beta}$.

1. Find the distribution of $\bar{X}$.
2. If $X_1, \dots, X_n$ are i.i.d. Exp(θ) with mean θ, find the distribution of $\sum_{i=1}^n X_i$.

Solution 1.7.

Part 1: By Theorem 2, for $t < \frac{n}{\beta}$,

$$M_{\bar{X}}(t) = \left(1 - \beta\frac{t}{n}\right)^{-n\alpha} = \left(1 - \frac{\beta}{n}t\right)^{-n\alpha},$$

which is the MGF of a $\Gamma(n\alpha, \frac{\beta}{n})$ random variable, so $\bar{X} \sim \Gamma(n\alpha, \frac{\beta}{n})$ by the uniqueness of MGFs. As a sanity check, $\mathbb{E}[\bar{X}] = n\alpha \cdot \frac{\beta}{n} = \alpha\beta$ and $\mathrm{Var}(\bar{X}) = n\alpha \cdot \frac{\beta^2}{n^2} = \frac{\alpha\beta^2}{n}$, which agrees with Theorem 1.

Part 2: The Exp(θ) distribution is the $\Gamma(1, \theta)$ distribution. The MGF of the sum is $[M_X(t)]^n = (1-\theta t)^{-n}$, so

$$\sum_{i=1}^n X_i \sim \Gamma(n, \theta).$$

Problem 1.8. Let X and Y be independent Unif$(0,1)$ random variables. Find the PDF of $Z = X+Y$ and of the sample mean $\frac{X+Y}{2}$.

Solution 1.8. By Theorem 3,

$$f_Z(z) = \int_{-\infty}^{\infty} \mathbb{1}(0 \le w \le 1)\mathbb{1}(0 \le z - w \le 1)\, dw = \int_{\max(0,z-1)}^{\min(1,z)} 1\, dw.$$

The integrand is non-zero only when $0 \le w \le 1$ and $z - 1 \le w \le z$, so

$$f_Z(z) = \begin{cases} z & 0 \le z \le 1 \\ 2 - z & 1 < z \le 2 \\ 0 & \text{otherwise.} \end{cases}$$

Since $\frac{Z}{2} \le x$ if and only if $Z \le 2x$, we have $F_{Z/2}(x) = F_Z(2x)$ and differentiating gives

$$f_{Z/2}(x) = 2f_Z(2x) = \begin{cases} 4x & 0 \le x \le \frac{1}{2} \\ 4(1-x) & \frac{1}{2} < x \le 1 \\ 0 & \text{otherwise.} \end{cases}$$

Even with only two observations, the sample mean is already more concentrated in the middle than a single uniform. This is the first step in the picture of the central limit theorem from Week 8.

Problem 1.9. Let $X_1, \dots, X_{10}$ be a random sample from N$(\mu, 4)$ where μ is unknown.

1. Compute $\mathbb{P}(S^2 > 6)$.
2. Find the constant c such that $\mathbb{P}(S^2 \le c) = 0.95$.

Solution 1.9. By Theorem **??**, $V = \frac{9S^2}{4} \sim \chi^2_9$, regardless of the value of μ.

Part 1: Using a χ^2 table (or software),

$$\mathbb{P}(S^2 > 6) = \mathbb{P}\left(\frac{9S^2}{4} > \frac{9 \cdot 6}{4}\right) = \mathbb{P}(V > 13.5) = 0.141.$$

Part 2: We have

$$0.95 = \mathbb{P}(S^2 \le c) = \mathbb{P}\left(V \le \frac{9c}{4}\right).$$

The 0.95 quantile of the χ^2_9 distribution is 16.919, so $\frac{9c}{4} = 16.919$, which gives $c = 7.52$.

Problem 1.10. Many integrals can be computed without integrating by recognizing a gamma PDF.

1. Compute $\int_0^\infty x^4 e^{-2x}\,dx$.
2. Compute $\int_0^\infty \sqrt{x}\, e^{-x/3}\,dx$.
3. Let $X \sim \Gamma(\alpha, \beta)$ with $\alpha > 1$. Compute $\mathbb{E}[\frac{1}{X}]$ and compare it with $\frac{1}{\mathbb{E}[X]}$.

Solution 1.10. Since the $\Gamma(\alpha, \beta)$ PDF integrates to 1, we have for all $\alpha, \beta > 0$,

$$\int_0^\infty x^{\alpha-1} e^{-x/\beta}\,dx = \Gamma(\alpha)\beta^\alpha. \tag{1}$$

Part 1: The integrand is $x^{5-1}e^{-x/(1/2)}$, so by (1) with $\alpha = 5$ and $\beta = \frac{1}{2}$,

$$\int_0^\infty x^4 e^{-2x}\,dx = \Gamma(5)\left(\frac{1}{2}\right)^5 = \frac{4!}{32} = \frac{3}{4}.$$

Part 2: The integrand is $x^{\frac{3}{2}-1}e^{-x/3}$, so by (1) with $\alpha = \frac{3}{2}$ and $\beta = 3$,

$$\int_0^\infty \sqrt{x}\, e^{-x/3}\,dx = \Gamma\left(\frac{3}{2}\right)3^{3/2} = 3^{3/2} \cdot \frac{\sqrt{\pi}}{2} = \frac{3\sqrt{3\pi}}{2} \approx 4.605.$$

Part 3: By Proposition 4 with $k = -1$ (which is allowed since $-1 > -\alpha$), and using $\Gamma(\alpha) = (\alpha-1)\Gamma(\alpha-1)$,

$$\mathbb{E}\left[\frac{1}{X}\right] = \frac{\Gamma(\alpha-1)}{\Gamma(\alpha)}\beta^{-1} = \frac{1}{(\alpha-1)\beta}.$$

This is larger than $\frac{1}{\mathbb{E}[X]} = \frac{1}{\alpha\beta}$, which is consistent with Jensen's inequality (Week 7) since $\frac{1}{x}$ is convex for $x > 0$.

Remark 9. Taking $\alpha = \frac{p}{2}$ and $\beta = 2$ in Part 3 gives $\mathbb{E}[\frac{1}{V}] = \frac{1}{p-2}$ for $V \sim \chi^2_p$, which is exactly what we needed for the variance of the t distribution (Problem **??**).

Problem 1.11. Let $X \sim \Gamma(n, \beta)$ where n is a positive integer.

1. Show that for all $t \geq 0$,

$$\mathbb{P}(X > t) = \sum_{k=0}^{n-1} e^{-t/\beta} \frac{(t/\beta)^k}{k!} = \mathbb{P}(N \leq n-1), \qquad \text{where } N \sim \text{Poi}(t/\beta).$$

2. Customers arrive at a coffee shop so that the times between arrivals are i.i.d. exponential random variables with mean $\frac{1}{2}$ minute (i.e. rate 2 customers per minute). What is the probability that the third customer arrives after more than 2 minutes?

Solution 1.11.

Part 1: Let $G(t)$ denote the sum on the right hand side. Both $\mathbb{P}(X > t)$ and $G(t)$ are equal to 1 at $t = 0$, so it suffices to show that they have the same derivative. Since X has PDF $f_X(t) = \frac{t^{n-1}e^{-t/\beta}}{(n-1)!\,\beta^n}$, we have $\frac{d}{dt}\mathbb{P}(X > t) = -f_X(t)$. Differentiating G with the product rule,

$$\begin{aligned} G'(t) &= -\frac{1}{\beta}\sum_{k=0}^{n-1} e^{-t/\beta}\frac{(t/\beta)^k}{k!} + \frac{1}{\beta}\sum_{k=1}^{n-1} e^{-t/\beta}\frac{(t/\beta)^{k-1}}{(k-1)!} \\ &= -\frac{1}{\beta}\sum_{k=0}^{n-1} e^{-t/\beta}\frac{(t/\beta)^k}{k!} + \frac{1}{\beta}\sum_{j=0}^{n-2} e^{-t/\beta}\frac{(t/\beta)^j}{j!}. \end{aligned}$$

The two sums telescope, and only the $k = n-1$ term of the first sum survives, so

$$G'(t) = -\frac{1}{\beta}e^{-t/\beta}\frac{(t/\beta)^{n-1}}{(n-1)!} = -\frac{t^{n-1}e^{-t/\beta}}{(n-1)!\,\beta^n} = -f_X(t).$$

Therefore, $\mathbb{P}(X > t) = G(t)$ for all $t \geq 0$.

Part 2: The interarrival times E_1, E_2, E_3 are i.i.d. $\text{Exp}(\frac{1}{2}) = \Gamma(1, \frac{1}{2})$, so by Proposition 5, the arrival time of the third customer is $T = E_1 + E_2 + E_3 \sim \Gamma(3, \frac{1}{2})$. By Part 1 with $\frac{t}{\beta} = \frac{2}{1/2} = 4$,

$$\mathbb{P}(T > 2) = e^{-4}\left(1 + 4 + \frac{4^2}{2!}\right) = 13e^{-4} \approx 0.2381.$$

Remark 10. Part 1 has a nice interpretation. If N counts the number of customers who arrive in the first t minutes, then $N \sim \text{Poi}(t/\beta)$ (customers arrive at rate $\frac{1}{\beta}$ per minute), and the nth customer arrives after time t if and only if at most $n-1$ customers arrived by time t. In other words, $\{X > t\} = \{N \leq n-1\}$.

1.6 Proofs of Key Results

Problem 1.12. Prove Proposition 1.

Solution 1.12. We construct both samples on the same probability space, a technique called *coupling*. Since the population values may repeat, we sample the indices instead of the values. Let $I_1, \ldots, I_K$ be i.i.d. uniform on $\{1, \ldots, N\}$ and set $Y_k = x_{I_k}$, so $(Y_1, \ldots, Y_K)$ is a sample with replacement. Let

$$D = \{I_1, \ldots, I_K \text{ are distinct}\}.$$

Step 1: Every ordered K-tuple of indices $(i_1, \ldots, i_K)$ has probability N^{-K} under sampling with replacement. Therefore, conditionally on D, the indices $(I_1, \ldots, I_K)$ are uniform over the $N(N -$

$1)\cdots(N-K+1)$ ordered K-tuples of distinct indices. This is exactly the distribution of the indices chosen when sampling without replacement, since each such tuple has probability $\frac{1}{N}\frac{1}{N-1}\cdots\frac{1}{N-K+1}$. Writing $\mathbf{X}=(X_1,\ldots,X_K)$ and $\mathbf{Y}=(Y_1,\ldots,Y_K)$, we conclude that for any set A,

$$\mathbb{P}(\mathbf{X}\in A)=\mathbb{P}(\mathbf{Y}\in A\mid D).$$

Step 2: By the law of total probability,

$$\begin{aligned}\mathbb{P}(\mathbf{Y}\in A)-\mathbb{P}(\mathbf{X}\in A)&=\mathbb{P}(\mathbf{Y}\in A\mid D)\,\mathbb{P}(D)+\mathbb{P}(\mathbf{Y}\in A,D^{\mathsf{c}})-\mathbb{P}(\mathbf{Y}\in A\mid D)\\&=\mathbb{P}(\mathbf{Y}\in A,D^{\mathsf{c}})-\mathbb{P}(\mathbf{X}\in A)\,\mathbb{P}(D^{\mathsf{c}}).\end{aligned}$$

Both terms on the right are between 0 and $\mathbb{P}(D^{\mathsf{c}})$, so their difference is at most $\mathbb{P}(D^{\mathsf{c}})$ in absolute value. We can estimate this probability by the union bound,

$$\mathbb{P}(D^{\mathsf{c}})=\mathbb{P}\left(\bigcup_{1\le i<j\le K}\{I_i=I_j\}\right)\le\sum_{1\le i<j\le K}\mathbb{P}(I_i=I_j)=\binom{K}{2}\frac{1}{N}=\frac{K(K-1)}{2N}.$$

Remark 11. The proof shows that the only difference between the two schemes is that sampling with replacement can pick the same element twice. If the population values are distinct, then taking A to be the set of vectors with a repeated entry shows that the difference is exactly $\mathbb{P}(D^{\mathsf{c}})$, which is the birthday problem probability.

Problem 1.13. Prove Proposition 2.

Solution 1.13.

Part 1: We add and subtract $\bar{x}$, just like in the proof of the best mean square estimator in Week 6, Problem 1.14,

$$\begin{aligned}\sum_{i=1}^{n}(x_i-a)^2&=\sum_{i=1}^{n}(x_i-\bar{x}+\bar{x}-a)^2\\&=\sum_{i=1}^{n}(x_i-\bar{x})^2+2(\bar{x}-a)\sum_{i=1}^{n}(x_i-\bar{x})+n(\bar{x}-a)^2.\end{aligned}$$

The cross term vanishes since $\sum_{i=1}^{n}(x_i-\bar{x})=n\bar{x}-n\bar{x}=0$. Therefore,

$$\sum_{i=1}^{n}(x_i-a)^2=\sum_{i=1}^{n}(x_i-\bar{x})^2+n(\bar{x}-a)^2\ge\sum_{i=1}^{n}(x_i-\bar{x})^2$$

with equality if and only if $a=\bar{x}$.

Part 2: Taking $a=0$ in the identity above gives $\sum_{i=1}^{n}x_i^2=\sum_{i=1}^{n}(x_i-\bar{x})^2+n\bar{x}^2$.

Problem 1.14. Prove Theorem 1.

Solution 1.14.

Mean: By linearity of expectation, $\mathbb{E}[\bar{X}]=\frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[X_i]=\frac{n\mu}{n}=\mu$.

Variance: By independence, the covariances vanish, so

$$\mathrm{Var}(\bar{X}) = \frac{1}{n^2}\sum_{i=1}^{n}\mathrm{Var}(X_i) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}.$$

Sample variance: By Proposition 2, $(n-1)S^2 = \sum_{i=1}^n X_i^2 - n\bar{X}^2$. Since $\mathbb{E}[Y^2] = \mathrm{Var}(Y) + \mathbb{E}[Y]^2$ for any random variable Y,

$$\mathbb{E}[X_i^2] = \sigma^2 + \mu^2 \qquad \text{and} \qquad \mathbb{E}[\bar{X}^2] = \frac{\sigma^2}{n} + \mu^2.$$

Therefore, by linearity

$$\mathbb{E}[(n-1)S^2] = n(\sigma^2+\mu^2) - n\left(\frac{\sigma^2}{n} + \mu^2\right) = (n-1)\sigma^2,$$

and dividing by $n-1$ gives $\mathbb{E}[S^2] = \sigma^2$.

Problem 1.15. Prove Theorem 2.

Solution 1.15. By the independence of the X_i,

$$M_{\bar{X}}(t) = \mathbb{E}\left[e^{\frac{t}{n}\sum_{i=1}^n X_i}\right] = \mathbb{E}\left[\prod_{i=1}^n e^{\frac{t}{n}X_i}\right] = \prod_{i=1}^n \mathbb{E}\left[e^{\frac{t}{n}X_i}\right] = \prod_{i=1}^n M_{X_i}\left(\frac{t}{n}\right) = \left[M_X\left(\frac{t}{n}\right)\right]^n,$$

where we used that the X_i are identically distributed in the last equality.

Problem 1.16. Prove Theorem 3.

Solution 1.16. We compute the CDF of Z. Since X and Y are independent, their joint PDF is $f_X(x)f_Y(y)$, so

$$F_Z(z) = \mathbb{P}(X+Y \le z) = \int_{-\infty}^{\infty}\int_{-\infty}^{z-x} f_X(x)f_Y(y)\,dy\,dx.$$

In the inner integral, we change variables to $u = x+y$, so $y = u-x$ and $dy = du$,

$$F_Z(z) = \int_{-\infty}^{\infty}\int_{-\infty}^{z} f_X(x)f_Y(u-x)\,du\,dx = \int_{-\infty}^{z}\left(\int_{-\infty}^{\infty} f_X(x)f_Y(u-x)\,dx\right)du,$$

where we used Fubini's theorem (the integrand is non-negative) to swap the order of integration. Since $F_Z(z) = \int_{-\infty}^z g(u)\,du$ for the non-negative function $g(u) = \int_{-\infty}^{\infty} f_X(x)f_Y(u-x)\,dx$, the function g is the PDF of Z.

Problem 1.17. Prove Proposition 3.

Solution 1.17.

Part 1: We integrate by parts with $u = x^\alpha$ and $dv = e^{-x}\,dx$,

$$\Gamma(\alpha+1) = \int_0^\infty x^\alpha e^{-x}\,dx = -x^\alpha e^{-x}\Big|_{x=0}^{x=\infty} + \alpha\int_0^\infty x^{\alpha-1}e^{-x}\,dx = \alpha\Gamma(\alpha).$$

The boundary terms vanish since $x^\alpha \to 0$ as $x \to 0$ (because $\alpha > 0$) and $x^\alpha e^{-x} \to 0$ as $x \to \infty$.

Part 2: We have $\Gamma(1) = \int_0^\infty e^{-x}\,dx = 1 = 0!$. If $\Gamma(n) = (n-1)!$, then Part 1 implies that $\Gamma(n+1) = n\Gamma(n) = n(n-1)! = n!$, so the result follows by induction.

Part 3: We use the substitution $x = \frac{z^2}{2}$ for $z > 0$, so $dx = z\,dz$ and $x^{-1/2} = \frac{\sqrt{2}}{z}$,

$$\Gamma\Big(\frac{1}{2}\Big) = \int_0^\infty x^{-\frac{1}{2}}e^{-x}\,dx = \int_0^\infty \frac{\sqrt{2}}{z}e^{-\frac{z^2}{2}}z\,dz = \sqrt{2}\int_0^\infty e^{-\frac{z^2}{2}}\,dz = \sqrt{2}\cdot\frac{\sqrt{2\pi}}{2} = \sqrt{\pi},$$

where we used that the $\mathrm{N}(0,1)$ PDF integrates to 1 and is symmetric, so $\int_0^\infty e^{-\frac{z^2}{2}}\,dz = \frac{1}{2}\sqrt{2\pi}$.

Problem 1.18. Prove Proposition 4.

Solution 1.18.

Part 1: By the definition of the expected value,

$$\mathbb{E}[X^k] = \frac{1}{\Gamma(\alpha)\beta^\alpha}\int_0^\infty x^{\alpha+k-1}e^{-x/\beta}\,dx.$$

Substituting $y = \frac{x}{\beta}$, so that $dx = \beta\,dy$,

$$\int_0^\infty x^{\alpha+k-1}e^{-x/\beta}\,dx = \beta^{\alpha+k}\int_0^\infty y^{\alpha+k-1}e^{-y}\,dy = \Gamma(\alpha+k)\beta^{\alpha+k},$$

which is finite since $\alpha + k > 0$. Therefore,

$$\mathbb{E}[X^k] = \frac{1}{\Gamma(\alpha)\beta^\alpha}\cdot\Gamma(\alpha+k)\beta^{\alpha+k} = \frac{\Gamma(\alpha+k)}{\Gamma(\alpha)}\beta^k.$$

Notice that the case $k = 0$ shows that the gamma PDF integrates to 1.

Part 2: By Part 1 and Proposition 3,

$$\mathbb{E}[X] = \frac{\Gamma(\alpha+1)}{\Gamma(\alpha)}\beta = \alpha\beta \qquad \text{and} \qquad \mathbb{E}[X^2] = \frac{\Gamma(\alpha+2)}{\Gamma(\alpha)}\beta^2 = (\alpha+1)\alpha\beta^2,$$

so

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = (\alpha^2+\alpha)\beta^2 - \alpha^2\beta^2 = \alpha\beta^2.$$

Part 3: For $t < \frac{1}{\beta}$, we have $e^{tx}e^{-x/\beta} = e^{-x/\tilde{\beta}}$ where $\tilde{\beta} = \frac{\beta}{1-\beta t} > 0$. The same computation as in Part 1 with $k = 0$ and β replaced by $\tilde{\beta}$ gives

$$M_X(t) = \frac{1}{\Gamma(\alpha)\beta^\alpha}\int_0^\infty x^{\alpha-1}e^{-x/\tilde{\beta}}\,dx = \frac{1}{\Gamma(\alpha)\beta^\alpha}\cdot\Gamma(\alpha)\tilde{\beta}^\alpha = \left(\frac{\tilde{\beta}}{\beta}\right)^\alpha = (1-\beta t)^{-\alpha}.$$

If $t \geq \frac{1}{\beta}$, then the integrand does not decay as $x \to \infty$, so the integral diverges.

Problem 1.19. Prove Proposition 5.

Solution 1.19.

Part 1: By Proposition 4, for $t < \frac{1}{c\beta}$,

$$M_{cX}(t) = \mathbb{E}[e^{tcX}] = M_X(ct) = (1 - \beta ct)^{-\alpha} = (1 - (c\beta)t)^{-\alpha},$$

which is the MGF of the $\Gamma(\alpha, c\beta)$ distribution, so $cX \sim \Gamma(\alpha, c\beta)$ by the uniqueness of MGFs.

Part 2: By the independence property of MGFs, for $t < \frac{1}{\beta}$,

$$M_{X_1+\cdots+X_k}(t) = \prod_{i=1}^{k} M_{X_i}(t) = \prod_{i=1}^{k} (1 - \beta t)^{-\alpha_i} = (1 - \beta t)^{-(\alpha_1+\cdots+\alpha_k)},$$

which is the MGF of the $\Gamma(\alpha_1 + \cdots + \alpha_k, \beta)$ distribution.

Remark 12. Problem 1.7 combines both parts. If $X_1, \ldots, X_n$ are i.i.d. $\Gamma(\alpha, \beta)$, then $\sum_{i=1}^{n} X_i \sim \Gamma(n\alpha, \beta)$ by Part 2, so $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \sim \Gamma(n\alpha, \frac{\beta}{n})$ by Part 1.